# Can the $n$=2 response function correct interference?

## 1 Setup

Consider a model with activation space $\mathbb{R}^D$ and response function

$$R(v) \;=\; F(S(v)), \qquad S(v) \;=\; \sum_{i=1}^{M} w_i \,(d_i^\top v)^2,$$

where $v \in \mathbb{R}^D$ is a perturbation to the activation, $F$ is convex and monotonic, $\{d_i\}_{i=1}^M$ are unit feature directions, and $w_i \geq 0$ are per-direction weights. Under superposition $M > D$.

## 2 Notation

- $D$: dimension of activation space.
- $M > D$: number of features.
- $d_i \in \mathbb{R}^D$, unit norm: direction of feature $i$.
- $\Phi \in \mathbb{R}^{D\times M}$: columns are $d_i$.
- $w_i \geq 0$: weight on direction $i$.
- $a_i$: activity of feature $i$. Inactive features have $a_i = 0$. Active features have $a_i$ of typical magnitude $\sigma_a$.
- $k$: number of simultaneously active features (sparsity). $k \ll D$.
- $G_{ij} = \langle d_i, d_j \rangle$: pairwise inner products.
- $c > 0$: threshold defining the plateau $\{v : S(v) < c\}$.

From $S(v) = \sum_i w_i (d_i^\top v)^2 = v^\top A v$ with $A = \sum_i w_i d_i d_i^\top \in \mathbb{R}^{D\times D}$, the plateau-breaking magnitude along unit direction $d$ is

$$m_d \;=\; \sqrt{c/(d^\top A\, d)}.$$

## 3 The question

When $k$ features are simultaneously active, an inactive feature $i$ has a spurious read-off $\hat{a}_i = d_i^\top h = \sum_{j \text{ active}} a_j G_{ij}$ of typical magnitude $\sigma_a \sqrt{k/D}$. Interference correction on feature $i$ requires this spurious read-off not to break the plateau, while genuine activation of feature $i$ (magnitude $\sigma_a$) does break it:

$$\sigma_a \sqrt{k/D} \;\leq\; m_{d_i} \;\leq\; \sigma_a.$$

We additionally require the model to be consistent with the linear representation hypothesis (LRH): downstream computation reads features through projections $d_i^\top h$, so the model must be

more sensitive along feature directions than along non-feature directions. For a random unit direction $d_r \in \mathbb{R}^D$:

$$m_{d_i} \;<\; m_{d_r} \qquad \text{for every feature } i.$$

This rules out isotropic $A$ (e.g. tight-frame packing with uniform weights), which would give equal sensitivity in every direction.

**Question.** Does there exist a choice of weights $\{w_i\}_{i=1}^{M}$ and threshold $c > 0$ such that both the per-feature correction condition and the LRH sensitivity condition hold simultaneously for every feature $i = 1, \ldots, M$?